

Although it revolutionized the computer and communications world like nothing before, the World Wide Web was only possible by bringing together several existing technologies: the telephone, television, radio, and Internet.

The Internet

The Soviet Union's launch of the *Sputnik 1* satellite in 1957 spurred the US Defense Department to consider means of communication after a nuclear attack. This led to the formation of the ARPANET (Advanced Research Projects Agency Network) in 1969, a system initially of four computers. In the mid-1980s, this growing network of interconnected computers became known as the Internet. Both the Internet and the World Wide Web were limited to academic and research organizations.

It wasn't until the 1993 launch of a user-friendly Web browser called Mosaic that the Web took off for more general use. Mosaic could show pictures as well as text, and users could follow Web links simply by clicking on them with a mouse. The Web became synonymous with the Internet, but they are distinct from one another. The World Wide Web facilitated navigation of the Internet and helped make the Internet such an effective mode of communication.

The computing revolution

The introduction in 1981 of IBM's 5150 personal computer drove a revolution in home and office computing. Smaller and cheaper than the large office computers, it and its successors had access to the Internet and email. With personal computers, the Internet saw huge growth. The first search engines began to appear in the early 1990s; Google, which is now almost synonymous with Web searches, arrived a little later, in 1997. The launch of online marketplace Amazon in 1994 revolutionized the way people shopped, allowing the purchase of everything from books and CDs to hotel rooms and airline tickets from the comfort of home.

The Internet brought about significant changes to the way businesses operated; globalization escalated, and the world



Sir Tim Berners-Lee, creator of the World Wide Web, was fascinated by computers from a young age. Today, he is an advocate for an open and free Internet.

seemed to become a much smaller place, with communication improved by the speed and efficiency of the Internet. Jobs were outsourced, and companies effectively became "nationless," since it was easier to operate from anywhere in the world.

The next wave of technological advances saw devices become smaller and more mobile due to electronic components on tiny integrated circuits, or "chips."

The future is now

Nowhere has the introduction of microchip technology had more impact than the introduction of the Apple iPhone in 2007. So-called smartphones have made the Internet a mobile resource, with wireless connectivity offering on-the-go access to news and satellite navigation, for example. Information and ideas can be shared from anywhere at the touch of a button via social-networking sites such as Facebook and Twitter. Smartphones have also had an impact on education, health care, and culture, and have changed the political landscape through use by protesters organizing rallies via social media to undermine regimes. Uprisings such as the Arab Spring, which began in 2010, were partly powered by activists who communicated across the Internet. Internet activism, or "clicktivism," has since become a powerful way to share ideas, raise awareness, or support a cause. With more than 3 billion users, the World Wide Web has transformed every aspect of modern daily life. ■